Day 2

From Genomes to Surveillance Action

Genomic Surveillance of Surgical Site Infections & AMR
08:30–17:00

Course files: github.com/aryakaul/ughe-ssi-genomics-course

Day 2 · Morning

Time	Session
08:30	Recap: your questions from yesterday
08:50	**Lab 4:** Who is it, and what does it resist?
10:30	*Break*
10:45	**Lecture 4:** Typing, trees and outbreaks
11:30	**Lab 5:** Pathogenwatch and Microreact
12:30	*Lunch*

Day 2 · Afternoon

Time	Session
13:15	**Lab 6:** Case study, a cluster on the surgical ward
15:00	*Break*
15:15	**Lecture 5:** Building a sustainable programme
16:00	Action planning
16:40	Knowledge check, evaluation, certificates

Your questions from yesterday

Answers to the exit tickets

> Today's files: github.com/aryakaul/ughe-ssi-genomics-course
> Lab 5–6 data are in `labs/data/`

Lab 4

Who Is It, and What Does It Resist?

Day 2 · 08:50–10:30 · 100 min
Pairs · swap driver and navigator at Part D

LO4: detect AMR genes and plasmids, type the isolate, and interpret the results alongside phenotypic AST

Open your handout

Lab 4 handout

`Lab4_species_typing_amr.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

The scenario

- A patient develops a **deep SSI 9 days** after an open fracture repair
- The lab reports *S. aureus*, **methicillin-resistant**
- The isolate was sequenced: you have its **assembly** (FASTA)

KEY MESSAGE

What can the genome add?

Teaching scenario. The genome is real public data from the Galaxy Training Network.

Plan for this lab

Part	What	Time
A	Get the genome	5 min
B	Who is it? Species and strain type	20 min
C	What does it resist? AMR genes	35 min
D	Is resistance mobile? Plasmids	15 min
E	Does it match the lab’s AST?	20 min

Get the genome

1. New history: `Lab4 – AMR`
2. **Upload** → **Paste/Fetch data** → paste the URL **from your handout**
3. **Type:** `fasta` · **Name:** `MRSA_contigs.fasta`

> **OPTIONAL**
>
> Your own Lab 2 assembly finished? Run every step on it too.

MLST: the strain type

Tool **MLST** → `MRSA_contigs.fasta` → scheme **automatic** → **Run**

```
MRSA_contigs.fasta   saureus   ST?   arcC(?) aroE(?) …
```

CHECK YOURSELF

1. Which **scheme** did MLST choose? What does it tell you?
2. What is the **ST**? Is it a well-known MRSA clone?

PART B2 · OPTIONAL

Species check with Kraken2

- Tool **Kraken2** on `MRSA_contigs.fasta`, with a **report**
- What is the top species?

> **KEY MESSAGE**
>
> **If the organism is wrong or the culture is mixed, every result after this is wrong.**

AMRFinderPlus (NCBI)

- **Input type:** nucleotide → `MRSA_contigs.fasta`
- **Organism:** `Staphylococcus_aureus` (finds point mutations too)
- Everything else: defaults → **Run** → open the **report**

KEY COLUMNS

Element symbol (gene) · **Class** (drug class) · **% Identity** · **% Coverage**

PART C2

staramr

- ResFinder + PointFinder + PlasmidFinder + MLST in one tool
- Input `MRSA_contigs.fasta` → defaults → **Run**
- Open **summary.tsv** and **detailed_summary.tsv**

PART C3

Record your findings

Gene / mutation	Drug class	AMRFinderPlus?	staramr?	On a plasmid?

Fill in the full table in your handout. The last column comes in Part D.

PART C

Check yourself: AMR genes

CHECK YOURSELF

1. Which gene explains the **methicillin resistance**? *(Glossary cheat sheet!)*
2. Did both tools find the same genes? If not, why might they differ?
3. What is staramr's **predicted phenotype**?

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

Is resistance mobile? Plasmids

1. In `detailed_summary.tsv`, find the **plasmid** rows (`rep…`) and note their **contig**
2. Mark resistance genes on the **same contig** as a plasmid

> **ALTERNATIVE**
>
> **ABRicate** with the **plasmidfinder** database

PART D

Check yourself: plasmids

CHECK YOURSELF

1. How many plasmid replicons? On which contigs?
2. Which resistance genes are probably on a **plasmid**? Why does that matter for infection control?
3. Why can’t we be **certain** from an Illumina assembly?

Does the genome match the lab's AST?

- Use the **AST report in your handout** (Part E)
- **It is hypothetical:** written for teaching, not the real AST for this genome
- For each antibiotic: which gene explains it? Do genome and AST **agree**?

PART E

Check yourself: genome vs AST

CHECK YOURSELF

1. Where do genome and AST **disagree**? Give two possible explanations for each.
2. What would you **tell the clinician**? Write one plain-language sentence.
3. Would you use the genome **instead of** AST to treat this patient?

What each tool told us

Tool	Answers
MLST	**Who?** Strain type, known clones
AMRFinderPlus / staramr	**What resistance?** Genes and mutations
PlasmidFinder	**Is it mobile?**
Comparing with AST	**How confident are we?**

Next: what about **many** isolates? → Lecture 4 and Lab 5

Extension for fast pairs

- **ABRicate** with the **card** and **vfdb** (virulence) databases
- Annotate the genome with **Bakta**
- Upload the FASTA to **ResFinder** on the web and compare

Adapted from the GTN tutorial “Identification of AMR genes in an assembled bacterial genome” (CC-BY 4.0).

Break

Back at 10:45 for Lecture 4

Lecture 4

Typing, Trees and Outbreaks

Day 2 · 10:45–11:30 · 45 min

LO5: type isolates and interpret a phylogeny to support or rule out an outbreak

“We’ve had 5 *Klebsiella* SSIs on Ward B in 6 weeks. Is this an outbreak?”

What is an outbreak?

- An outbreak means **transmission**: the same strain passing between patients
- Directly, or through staff, equipment or the environment

KEY MESSAGE

Same species ≠ same strain.
Same AST pattern ≠ same strain.

Typing: naming bacteria at different resolutions

Method	Like a…	Used for
Species	country	Clinical ID
MLST (7 genes) → ST	city	Is it a known high-risk clone?
cgMLST (1,000–3,000 genes)	street	Clusters within an ST
SNPs (whole genome)	house number	Outbreaks and transmission

MLST in one slide

- Seven “housekeeping” genes; each version gets a number
- The combination of 7 numbers = the **Sequence Type (ST)**
- Known high-risk clones: *K. pneumoniae* ST258, ST307 · MRSA ST8, ST22
- Tools do it automatically (Lab 4, Lab 5)

KEY MESSAGE

Same ST = possibly related. Different ST = not a recent outbreak.

SNP distances

- Align genomes to a reference
- Count the positions where they differ
- **Few SNPs** → recent common ancestor → possible transmission
- **Many SNPs** → not recent transmission

ROUGH GUIDE (VARIES BY STUDY)

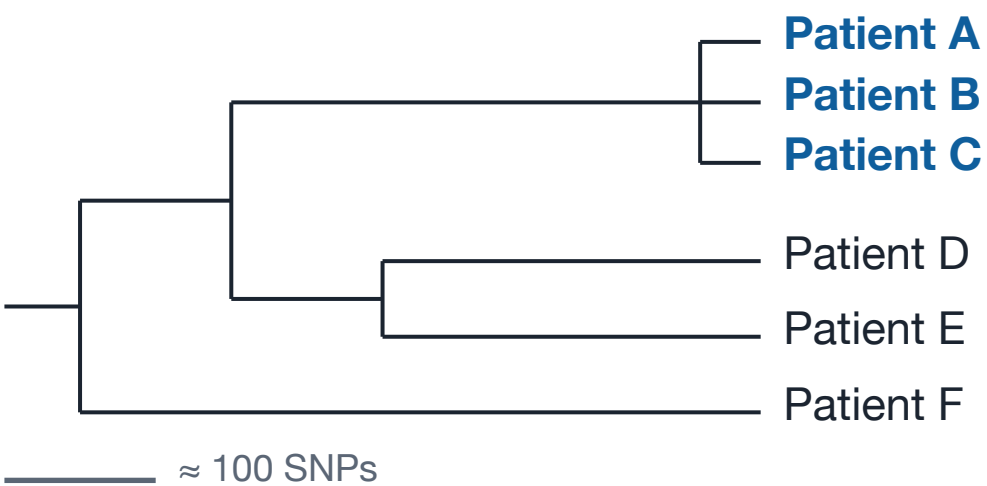
S. aureus: **≤ 15–25 SNPs**

K. pneumoniae, *E. coli*: **≤ 10–25 SNPs**

KEY MESSAGE

Thresholds are guides, not laws. Always combine them with epidemiology.

Reading a phylogenetic tree



- **Tips** = isolates
- **Branch length** = genetic change
- Tips on **short branches** close together are closely related (A, B, C)

COMMON MISTAKES

- Reading top-to-bottom order as meaningful
- Ignoring the **scale bar**

Genomics + epidemiology: time, place, person

A genomic cluster is an outbreak only if the **epidemiology fits**: same ward or theatre at overlapping times? Same surgeon, equipment or staff?

Confirm
Same strain, few SNPs, overlapping stays

Refute
Different STs: no need to close the ward

Find hidden links
Cases on different wards linked by shared equipment or staff

Strain outbreak vs plasmid outbreak

STRAIN OUTBREAK

The **same clone** in many patients

Tree: one tight cluster

PLASMID OUTBREAK

Different strains or species, but the **same resistance plasmid**

Needs Nanopore or hybrid data to confirm

KEY MESSAGE

Tree says “unrelated” but the resistance is unusual and identical? Think plasmid.

Vote: outbreak or not? (cards A / B / C)

1. **4 MRSA**, all ST22, 2–5 SNPs apart, same ward over 3 weeks
2. **4 *Klebsiella***: ST15, ST307, ST14, ST101
3. **3 *E. coli***, all ST131, 80–150 SNPs apart, 3 different hospitals

Vote: answers

1. **Likely outbreak**
2. **Not a single-strain outbreak.** Check for a shared plasmid?
3. **A common circulating clone**, not recent transmission

From result to action

Finding	Possible IPC action
Confirmed cluster	Enhanced cleaning, cohorting, contact screening, theatre audit, staff screening if indicated
Refuted cluster	Avoid unnecessary ward closure; look at other causes (e.g. prophylaxis timing)
High-risk clone or new carbapenemase	Notify the national reference lab / public-health authority

> Communicate in **plain language** to clinicians and managers. You'll practise this in Lab 6.

Tools for the next lab

PATHOGENWATCH

Upload assemblies → species, ST, AMR, clustering, tree

MICROREACT

Tree + map + timeline + metadata in one interactive view

Both are free, used by national public-health labs, and need **no coding**.

Lab 5

Pathogenwatch and Microreact: Many Genomes at Once

Day 2 · 11:30–12:30 · 60 min
Pairs

LO5: use free web platforms to type, screen and visualise a collection of genomes

Open your handout

Lab 5 handout

`Lab5_pathogenwatch_microreact.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

The scenario

- The IPC nurse at **District Hospital A** is worried about more **MRSA SSIs on Surgical Ward B**
- The lab stored the isolates and sent them for sequencing
- You have **genomes from all MRSA isolates of the past months**

KEY MESSAGE

Get a quick overview of all the genomes at once.

The genomes are real public data. The hospital, wards and dates are fictional.

Plan for this lab

Part	What	Time
A	Pathogenwatch: typing and AMR	30 min
B	Microreact: tree + timeline	25 min
	Summary	5 min

Files from the instructor

File	Contains
`assemblies/`	One FASTA per isolate
`koser_tree.nwk`	Phylogenetic tree (Galaxy: Snippy → IQ-TREE)
`lab6_metadata.csv`	Patient, ward, procedure and date information

- **GitHub:** github.com/aryakaul/ughe-ssi-genomics-course
- Green **Code** button → **Download ZIP** → unzip → `labs/data/`

Upload and explore

1. Log in to `pathogen.watch` → **Upload** → drag in the FASTA assemblies
2. Wait for each genome to be analysed (a few minutes)
3. For **at least 6** genomes, record:

Genome	Species	ST	Predicted resistances	QC pass?

Does the organism agree with the lab? Check the assembly QC metrics (length, contigs, N50).

Make a collection and view the tree

1. Select all your genomes → **Create collection** → name it `UGHE Lab 5`
2. Open the collection → explore the **tree** view
3. Explore the **AMR** view: resistance profiles side by side

CHECK YOURSELF

1. How many different STs are in the collection?
2. Do all isolates have the **same** AMR profile? What does that suggest, and what does it **not** prove?
3. In the tree, can you see a group of isolates on very **short** branches?

Create a project

1. Go to `microreact.org` → **Upload** (sign in only if you want to save)
2. Drag in **both** files: `lab6_metadata.csv` and `koser_tree.nwk`
3. Column that links the table to the tree tips → **`id`**
4. Dates: confirm that **`year`**, **`month`**, **`day`** form the date
5. **Continue / Create**

> “No matching IDs”? Raise your hand: the tree labels and the `id` column must match exactly.

Explore: tree + time + place

- **Colour** the tree by `ward`, then `theatre`, then `hospital`
- Open the **Timeline**: when did each infection happen?
- **Click a tip**: its row lights up in the table and timeline
- **Lasso-select** a group of close tips: where and when are they?

CHECK YOURSELF

1. Is there a group that is closely related **and** close in time **and** in the same place?
2. Is any isolate from the **same ward at the same time not** in that group?
3. Is any isolate in the group from a **different ward**? What could link it?

PART B3

Save, share, summarise

Save (if signed in) and copy the project link. **You'll use it in Lab 6.**

Tool	Gives you	Input
Pathogenwatch	Species, ST, AMR, QC, clustering, global context	FASTA assemblies
Microreact	Visual story: tree + time + place + metadata	Tree (Newick) + table (CSV)

Both are free and used by national public-health labs. **You can use them on Monday with your own data.**

Pathogenwatch: Centre for Genomic Pathogen Surveillance. Microreact: Argimón S, et al. Microb Genom 2016.

Lunch

Back at 13:15 for the case study (Lab 6)

Lab 6

Case Study:
A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min
Groups of 4

LO3–LO5: combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

Open your handout

Lab 6 handout

`Lab6_ssi_outbreak_case_study.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

About this case

- The **genomes are real**: MRSA isolates from a published outbreak investigation
- The **hospitals, wards, patients and dates are fictional**, so the case feels like our setting

> The real story is revealed at the end.

THE SITUATION

District Hospital A · Monday, 16 March 2026

The IPC nurse writes to you:

> *“Since early February we’ve had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it’s bad luck. Theatre staff say it’s the ward. The ward says it’s the theatre.*

The question

> *The director asks whether we should **close Ward B**. That would cancel about 40 operations a week. We got genomes back last week for all MRSA isolates since September.*
>
> ***Can you tell us what's going on by Wednesday's IPC meeting?"***

Your evidence

1. **Microreact project** from Lab 5: tree + timeline + metadata
2. **SNP matrix:** `koser_snp_matrix.tsv` (opens in Excel)
3. **Pathogenwatch results** from Lab 5: ST and AMR
4. **Metadata:** `lab6_metadata.csv`

Group roles

Role	Job
Genomics lead	Reads the tree and SNP matrix
Epidemiologist	Builds the timeline: who was where, when
Clinician / IPC	Thinks through transmission routes and actions
Communicator	Writes and gives the 5-minute briefing

Plan for this lab

Part	What	Time
A	Describe the cases	15 min
B	What does the genome say?	25 min
C	Genomics + epidemiology together	20 min
D	Prepare the briefing	25 min
E	Briefings to the IPC committee	15 min

Describe the cases

Sketch an **epidemic curve**: weeks across, number of cases up.
Two colours: **Ward B** and **other wards / hospitals**.

QUESTIONS

1. How many MRSA isolates in total? How many from Ward B since 1 February?
2. Do all Ward B cases look the same on the **lab report**? Does that make them one outbreak?

What does the genome say?

1. **Sequence types:** group the isolates by ST (Pathogenwatch)
2. **SNP distances:** highlight all pairs **≤ 15 SNPs** apart
3. Which isolates form a **cluster**, all close to each other?
4. **Tree:** colour by **ward**, then by **theatre**. Is the cluster a tight group on short branches?

≤ 15 SNPs: a common working threshold for possible recent *S. aureus* transmission

Put genomics and epidemiology together

For **each** Ward B case since February:

Patient	In the cluster?	Ward / theatre / time link?	Linked · not linked · uncertain

Use the full table in your handout.

PART C

Discuss in your group

1. Is there an outbreak? How confident are you?
2. Is a Ward B patient **outside** the cluster, though on the ward at the same time?
3. Is a cluster patient **not** on Ward B? What do they share? → **likely route?**
4. Are isolates from **other hospitals** in the cluster?
5. What **other information** would you want?

Prepare a 5-minute briefing

For the IPC committee: director, chief surgeon, nursing lead, lab head. **Not genomics experts.**

RULES

- One flip-chart page or 3 slides
- Explain any jargon: “SNP” → “genetic differences”
- **No more than one figure**

PART D

Briefing template

1. **Bottom line:** “We found / did not find evidence of an outbreak of ___ linked to ____.”
2. **What we did:** “We sequenced __ MRSA isolates from __ to __.”
3. **What we found:** one visual + 2–3 bullets
4. **What it means:** which cases are linked; the likely route
5. **Recommendations:** 3 actions, with who and by when
6. **What we don’t know yet**

Briefings to the IPC committee

Each group has **5 minutes**. The committee must ask at least one question:

“Should we close Ward B?”

“Is it the surgeons' fault?”

“How sure are you?”

“What will this cost?”

Pause

Instructor: don’t go to the next slide until every group has given its briefing

The reveal

The genomes come from **Köser et al., *NEJM* 2012**: a suspected MRSA outbreak in a **neonatal unit** in Cambridge, UK.

- Sequencing separated the **outbreak isolates** from unrelated MRSA that looked identical on routine tests
- A follow-up study (**Harris et al. 2013**) traced transmission into the community and found a **staff carrier**, which ended the outbreak

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.

Take-home messages

1. **Same species + same antibiogram ≠ same outbreak**
2. Genomics finds **hidden links**: shared theatre, staff, equipment
3. Genomics can **prevent unnecessary actions**, like closing a ward
4. Genomics is only useful with **good metadata** and **clear communication**

Break

Back at 15:15 for Lecture 5

Lecture 5

Building a Sustainable Genomic Surveillance Programme

Day 2 · 15:15–16:00 · 45 min
≈ 25 min talk + 20 min discussion

LO6: design a sustainable, outsourced genomic surveillance workflow for your own setting

What we can do on Monday

Collect → **Store** → **Send** → **Analyse** → **Act**

No sequencer is needed to start. We need:

- a **plan** and a clear question
- **stored isolates** with good **metadata**
- a sequencing **partner**
- **people** who can analyse the data

Choose a question first

Type	Example	What to sequence
Outbreak (reactive)	“Is this ward cluster one strain?”	All cluster isolates + a few unrelated controls
Routine (proactive)	“Which ESBL/ carbapenemase genes cause our SSIs?”	e.g. all resistant SSI isolates for 6 months
Research	“Are post-caesarean SSI *E. coli* from the community?”	A designed sample with matched community isolates

Start small.

50–100 well-chosen isolates with complete metadata beat 1,000 random ones.

Sampling and biobanking

- Store **every** SSI isolate with resistance of interest (at least ESBL, carbapenem-resistant, MRSA)
- –80 °C freezer or glycerol stocks; a **backup** at a second site if possible
- Keep an **isolate register** (spreadsheet): study ID, organism, date, ward, specimen, AST, freezer location
- Agree on labelling **before** you start

Metadata and data management

- Use a template aligned with **PHA4GE** fields
- **Coded IDs only.** Keep the ID → patient key secure and separate
- Name files consistently: `UGHE-SSI-0001_R1.fastq.gz`
- Back up raw reads in **two places**: they can't be replaced
- Galaxy histories are a free, reproducible record of every analysis

Ethics, agreements and data sharing

- **Surveillance and IPC:** usually covered by routine public-health practice
- **Research:** needs ethics approval (Rwanda National Ethics Committee + institutional IRB)
- **Sending isolates abroad:** material transfer agreement (MTA)
- **Data-sharing agreement:** who owns the data and how it is published
- **Share genomes publicly** (ENA / NCBI) with non-identifying metadata

KEY MESSAGE

Equity: local investigators should lead the analysis and authorship.

Partners in the region

RBC
Rwanda Biomedical Centre / National Reference Laboratory

Africa CDC PGI
Regional sequencing and training network

ASLM
AMR surveillance programmes such as MAAP

Other partners
Universities, NGOs, Fleming Fund projects, commercial services

Ask the room: which of these do we already work with?

Reporting results so they change practice

Audience	What they need
Clinicians	“This isolate carries an NDM carbapenemase. Avoid carbapenems; consult ID.”
IPC committee	Cluster summary, tree, timeline, recommended actions
Ministry / RBC	Quarterly summary of resistance genes and clones

> Use simple visuals (Microreact) and a one-page template. **Close the loop:** did the action reduce SSIs?

People: keeping the skills alive

- **2–3 “genomics champions”** per department
- A monthly **data clinic / journal club** for course graduates
- Self-paced learning: Galaxy Training Network, The Carpentries, Africa CDC, Wellcome Connecting Science
- Build genomics into **student teaching** (MGHD, MBBS research projects)

Funding

- Build sequencing costs into grants (AMR, surgery, maternal health)
- Funders active in Africa: Fleming Fund, Wellcome, Gates Foundation, Africa CDC, NIH / Fogarty, EDCTP

BUDGET PER ISOLATE

Sequencing + shipping + staff time + data storage

Your one-page action plan

1. **Question:** what do I want to know?
2. **Isolates:** which, how many, from where, when? Already stored?
3. **Metadata:** which fields? Who collects them?
4. **Partner:** who? Which platform? Cost? Timeline?
5. **Analysis:** which tools? Who does it?
6. **Ethics:** IRB? MTA? Data sharing?
7. **Communication:** who gets results, in what format?
8. **First step next week:** _____

Discussion

20 min · small groups, then the whole room

1. What is the **single biggest barrier** to starting genomic surveillance at our institution? How could we overcome it?
2. Which **question** (outbreak, routine, research) would be most valuable to answer first?

Action planning

Draft **your own one-page plan** for a genomic surveillance mini-project.

- Use the 8-point template from Lecture 5
- Choose a question you could really start on next week
- Be concrete: **which** isolates, **which** partner, **who** does what

KEY MESSAGE

Last point on the template: your first step next week.

Knowledge check and evaluation

- The same knowledge check as yesterday
- Use the **same code word** and tick **Post**
- Answer **N4** at the end: what support would help you most?
- Then the course evaluation

Keep going

- **Course files:** github.com/aryakaul/ughe-ssi-genomics-course
- Your **Galaxy histories** stay in your account: re-run them on new data
- Next steps: `resources/further_learning.md`
- A monthly **data clinic / journal club** for course graduates

Thank you!

Certificates now